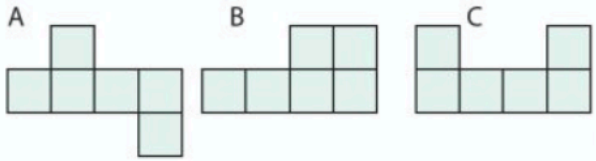
Exercise 5G

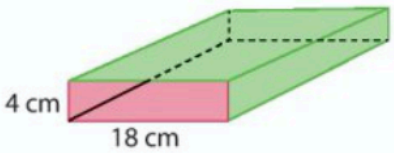
1. Examine the nets over.
Which of these are possible nets of a cube?
Give a reason for your answer.

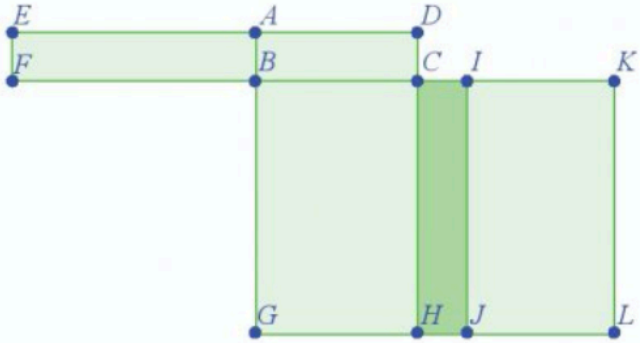


2. Draw a net for a rectangular solid with dimensions 20 cm × 5 cm × 2 cm and hence find the surface area of the solid.

3. Draw a net for a hollow cylinder of height 10 cm and radius 7 cm.
Using the π value on your calculator, find the curved surface area of the cylinder correct to 3 significant figures.

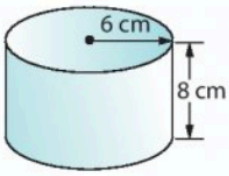
4. An open (red end) rectangular box has dimensions 4 cm × 18 cm × 20 cm as shown.
A net for this box is given below.
Using the diagram, copy and complete the chart by finding the 2 dimensions for each flat surface named in the chart.
Hence find the total (external) surface area of the box.





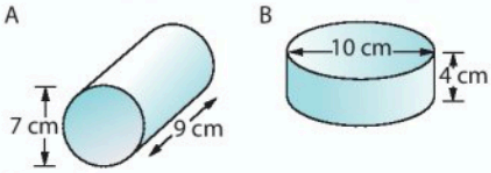
Face	Dimensions	Area
ABCD	( ) × ( )	
BCGH	( ) × ( )	
EFAB	( ) × ( )	
HJIC	( ) × ( )	
IKJL	( ) × ( )	
	Total =	

5. A solid cylinder has the dimensions as shown:



(i) What is the radius of the cylinder?
(ii) What is the height of the cylinder?
(iii) Calculate the circumference of the cylinder.
Use the π value on your calculator and give your answer correct to 1 decimal place.
(iv) Draw a net for this cylinder writing in the dimensions obtained from parts (i) ,(ii) and (iii).
(v) Use this net to find the total surface area of the solid cylinder.

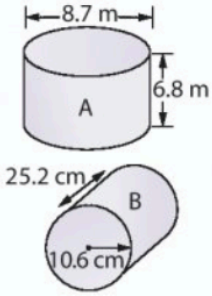
6. By studying the diagrams over



Write down: (i) radius_A
(ii) height_A
(iii) radius_B
(iv) height_B

Use these measurements to find the total surface area of A and B.
Take $\pi = 3.14$ and give each answer correct to 1 decimal place.

7. For cylinder A, find:

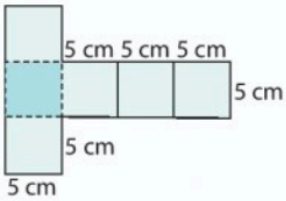


(i) the radius of A
(ii) the curved surface area of A
(iii) the area of the circular ends
(iv) the surface area of A

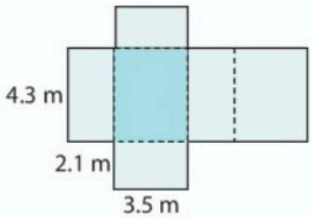
For cylinder B, find:
(i) the curved surface area
(ii) the total surface area

Take $\pi = 3.14$ and give each answer correct to 3 significant figures

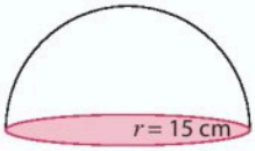
8. Using the net find the surface area of the figure shown.
If this net is folded along the dotted lines describe the figure produced.



9. A rectangular box has a net as shown find its surface area. Give your answer correct to the nearest whole number.
Two similar boxes made from cardboard are needed. Using squared paper draw a diagram of a net of the two boxes so that the least amount of cardboard is needed.



10. Calculate the surface area of a sphere of radius 14 cm.
Take $\pi = 3.14$ and give each answer correct to 2 significant figures.

11. The surface area of a sphere is 5 544 cm^2.
Calculate the radius of the sphere. Take $\pi = \frac{22}{7}$

12. Write down the formula for the surface area of a sphere.
Hence work out a formula for the surface area of a solid hemi-sphere.
Using this formula find the surface area of a hemi-sphere of radius 15 cm.
Take $\pi = 3.14$ and give each answer correct to 2 places of decimals.